



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

Re - accredited by NAAC with 'A' Grade

Founder: Prof. Dr. S. B. Mujumdar, M.Sc., Ph.D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name : Basic Electrical and Electronics Engineering
Course Code : T7540
Faculty : Engineering
Course Credit : 3
Course Level : 1
Sub-Committee (Specialization) : Electronics and Telecommunication
Learning Objectives :

- The student will be able to:
- Use Kirchhoffs laws to analyse DC circuits.
- Explain the working of electrical measuring instruments.
- Describe the working principle of DC Generator and Motor.
- Illustrate the basic principles of Electromagnetic Induction and its applications such as Transformer and Induction Motors.
- Demonstrate the working of different semiconductor devices and their applications.
- Discuss different different transistor configurations and compare them on the basis of current gain, voltage gain, input/output impedance..
- Represent numerical values in various number systems and perform number conversions between different number systems.
- Demonstrate the operation of logic gates.
- Implement Boolean expressions using logic gates.
- Implement half adder, full adder using logic gates.

Books Recommended :

Book	Author	Publisher
A Textbook of Electrical Technology - Volume - I	B. L. Theraja	S. Chand & Company Ltd., New Delhi.
Basic Electrical Engineering	V. K. Mehta,	S. Chand & Company Ltd., New Delhi
Digital Fundamentals	Thomas L. Floyd	UBS Publishers
Electronic Devices & Circuits	Allen Mottershed	PHI.
Electronic devices and Circuits Theory	Robert L. Boylestead, Louis Nashelsky.	PHI
Electronic Principles	Malvino	PHI.
Fundamentals of Logic Design	Charles H. Roth	JAICO Publishing House.
Issues in Power diode modelling for circuit simulation	International conference on power electronics and drives, Singapore	NA
Simulation of R-L-C series and parallel resonance in basic electrical engineering with labview	Research Journal of Engineering Sciences	NA
Theory & Problems of Basic Electrical Engineering	I.J. Nagrath & Kothari	Prentice-Hall of India Pvt. Ltd.

Course Outline :

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence

1	Electric circuits and Measurements Ohm's law Kirchoff's laws Steady state solution of DC circuits Introduction to AC circuits waveforms and RMS value power and power factor Single phase and three phase balanced circuits. Operating principles of moving coil and moving iron instruments dynamometer type watt meters and energy meters.	17	17
2	Electrical Machines Construction Principle of operation basic equations of DC generators DC motors single phase transformer and single phase Induction motor.	8	8
3	Semiconductor Devices and Applications Characteristics of PN junction diode Zener effect Zener diode and its characteristics Half wave and full wave rectifiers Voltage regulation. Bipolar junction transistor CB,CE, CC configurations and characteristics elementary treatment of small signal amplifiers.	14	14
4	Digital Electronics Binary and Hexadecimal number system logic gates Boolean algebra Half and full adders.	6	6
Total		45	45

Pre Requisites :

None.

Evaluation :

Assignment
 Quiz
 Class test
 Seminar
 Examination

Pedagogy :

Interactive classroom teaching.
 Problem based learning